Research Methodology

Research Question to Protocol

Guide to Systematic Literature Review

By:

Dr. Zohair Ahmed

Turn complex literature reviews into manageable steps!

- Writing a systematic literature review (SLR) isn’t just about summarizing articles, it's about following a clear, structured process to gather and synthesize the best available research on a specific question.
- Unlike traditional reviews, an SLR is designed to be transparent, reproducible, and comprehensive. It involves carefully planning every step, from formulating a research question to extracting and analyzing data.

Define the Research Question

- Every strong systematic review starts with a well-defined question.
- This shapes the direction of your entire review, from what studies you search to how you interpret findings.
- Use frameworks like **PICO** (Population, Intervention, Comparison, Outcome) or **SPIDER** (Sample, Phenomenon of Interest, Design, Evaluation, Research Type) to give structure to your question.
- **Example (PICO):**
- How does mindfulness-based therapy (Intervention) compare to traditional cognitive behavioral therapy (Comparison) in reducing anxiety (Outcome) among adults (Population)?
- A focused question ensures your review is targeted, relevant, and methodologically sound.

Define the Research Question

- it’s not necessary (or even always appropriate) to include every **PICO** element in every research question for a CS/AI SLR.
- Use PICO as a guide, not a straitjacket. In computer science, **“Comparison”** is often optional, and many reviews use adapted frameworks **like PICOC (adds Context)** or **PICOS (adds Study design)** so you can tailor elements per question.

How to decide which elements to include

- **Descriptive/mapping questions (what has been done?)**
  - Use: P (Population), I (Intervention), Ctx (Context)
  - Often omit: C (Comparison), O (Outcome) unless you summarize results
  - **Example:** “What model architectures have been used for sentiment analysis on social media?”
  - **P:** social media SA datasets; **I:** model architectures; **C: —; O: —; Ctx:** social media
- **Effectiveness/comparative questions (what works better?)**
  - Use: P, I, C, O (+ Ctx if relevant)
  - **Example:** “Do transformer-based models outperform classical ML for sentiment analysis in low-resource languages?”
  - **P:** SA datasets in low-resource languages; **I:** transformers; **C:** classical ML; **O:** F1/accuracy; **Ctx:** low-resource NLP
- **Efficiency/resource questions (costs, latency, memory)**
  - Use: P, I, O (+ C if you compare), Ctx if hardware matters
  - **Example:** “What is the inference latency of BERT-like vs. distilled models on mobile devices?”
  - **P:** SA workloads; **I:** BERT-like vs distilled; **C:** implicit between models; **O:** latency/memory; **Ctx:** mobile/edge

Recommended frameworks for CS/AI SLRs

- PICO: Population, Intervention, Comparison, Outcome
- PICOC: Adds Context. Very useful in CS (e.g., platform, domain, language, data source).
- PICOS: Adds Study design. Useful if you only include empirical/experimental studies.
- SPIDER: For qualitative syntheses; less common for mainstream NLP benchmarking SLRs.

Should Merge Two Question in to One
(“models” and “datasets”)

- Keep models and datasets as separate research questions.
- Don’t merge them into one.
- Then make the **“which model outperforms”** question explicitly **dataset** and **metric-specific** to avoid apples-to-oranges comparisons.
- **Why separate “models” and “datasets”**
  - Combining them bloats one question and makes synthesis messy. Each topic is large enough to warrant its own mapping.
  - Performance rankings are only meaningful within the same dataset/split/metric; separating lets you structure RQ3 properly.
  - It simplifies search strings, screening, and extraction (one set of fields for models; another for datasets/evaluation).

Recommended three-question set (refined)

- **RQ2:** Datasets and evaluation (PICOC/PICOS): What datasets and evaluation setups exist for [Task] in [Context]?
- P: [Task]
- I: Datasets/resources and evaluation protocols (splits, label schemes)
- C: N/A
- O: N/A (or “reported metrics” as descriptive)
- Ctx: Domain, language, time period
- S (optional): Empirical benchmark studies

Recommended three-question set (refined)

- **RQ3:** Effectiveness (PICO): Among comparable studies on the same dataset/split and metric, which models achieve the best performance, and under what conditions?
- P: Studies evaluating [Task] on specified datasets
- I: Candidate model families
- C: Other models/baselines on the same dataset/split
- O: Standardized metrics (prefer macro-F1 over accuracy for imbalance; also, AUROC, F1 per class)
- Ctx: Domain/language; note pretraining, compute budget, and parameter count

Recommended three-question set (refined)

- **RQ1:** Models (PICOC): What modeling approaches have been applied to [Task] in [Context], and how can they be categorized?
- **P:** [Task data, e.g., sentiment text]
- **I:** Modeling approaches (e.g., lexicon-based, SVM/CNN/RNN, transformers, prompting, adapters)
- **C:** N/A
- **O:** N/A
- **Ctx:** Domain/language/platform (e.g., product reviews, Twitter/X, healthcare; mono/multilingual)

Recommended three-question set (refined)

- ("sentiment analysis" OR "opinion mining" OR "polarity classification")
- AND (Twitter OR "social media" OR "TweetEval" OR "Sentiment140")
- AND (benchmark OR dataset OR evaluation OR accuracy OR "macro-F1")
- AND (2018..2025)
- **Data extraction form (columns)**
  - StudyID, Year, Venue
  - RQ tags (RQ1/RQ2/RQ3)
  - Model family, variant, params
  - Dataset, domain, language, size, label scheme
  - Split (official/dev/test), preprocessing
  - Metric(s) and value(s) (+ variance if given)
  - Resources (code/data), hardware/latency if reported

SLR title that aligns with your three RQs

- (Models; Datasets & Metrics; Comparative)
- Key things to include
  - Task and level: “Sentiment Analysis” (document/sentence/aspect-based).
  - Context: domain (social media, reviews, healthcare), language(s), modality if relevant.
  - Evidence type: “Systematic Literature Review” (add “and Meta-Analysis” only if you’ll synthesize effect sizes).
  - Time window: years covered (optional but helpful).
  - Outcomes focus: if RQ3 is central, signal “Comparative Performance,” “Accuracy,” or “Evaluation.”
- Avoid
  - Jargon like PICO in the title.
  - Overlong lists; use a concise main title + informative subtitle.
  - Overpromising (e.g., say “Meta-Analysis” only if you do it).

Title formulas (pick one and fill)

- Models, Datasets, and Comparative Performance for [Task] in [Context]: A Systematic Literature Review ([Years])
- [Task] in [Context]: Models, Datasets, and Evaluation—A Systematic Literature Review ([Years])
- [Task]: Methods, Datasets, and Accuracy Benchmarks in [Context]—Systematic Review ([Years])
- If you're mostly mapping: [Task] in [Context]: Methods, Datasets, and Evaluation—A Systematic Mapping Study

Develop a Protocol

- Think of the protocol as your SLR blueprint. It outlines exactly how you'll conduct your review, reducing the risk of bias or inconsistency.
- Your protocol should include:
  - Research question and objectives
  - Inclusion and exclusion criteria
  - Databases and search strategies
  - Screening process
  - Data extraction and synthesis methods

Conduct a Comprehensive Literature Search

- This step is all about coverage and precision.
- Search multiple databases: PubMed, Scopus, Web of Science, Google Scholar, to capture relevant studies. Use Boolean operators (AND, OR, NOT), MeSH terms, and filters to fine-tune your results.
- Example search query:
  (“physical activity” OR “exercise”) AND (“mental health” OR “depression”) AND (“adolescents” OR “teenagers”)
- Document every step, databases used, date ranges, and search strings, so your review can be replicated by others.
- **Use Given Google Sheet**

Select Studies Based on Eligibility Criteria

- To stay objective, use predefined inclusion and exclusion criteria.
- Screen in two phases:
  - Title and abstract screening
  - Full-text review
- Use a PRISMA flow diagram to visually track the number of studies included and excluded at each stage.
- Example criteria:
  - Inclusion: Peer-reviewed articles from 2013–2023, English language, randomized controlled trials
  - Exclusion: Editorials, opinion pieces, non-human studies
- This step ensures your final pool of studies is both relevant and high quality.

Extract Relevant Data

- This step is about systematically capturing essential study details.
- Use a structured data extraction form or spreadsheet that includes:
  - Study title and year
  - Authors and affiliations
  - Sample size and demographics
  - Intervention or exposure details
  - Measured outcomes
  - Key findings or effect sizes
- Tools like Covidence or RevMan can make extraction faster and more organized.

Analyze and Synthesize Findings

- Once the data is in, it's time to make sense of it all.
- You can:
  - Use narrative synthesis for qualitative reviews—identify patterns and themes across studies
  - Conduct a meta-analysis for quantitative reviews—statistically combine results
- Focus on:
  - Comparing outcomes
  - Highlighting contradictions
  - Weighing results by study quality

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- Example (narrative synthesis):

  Several studies found that mindfulness interventions reduced anxiety among youngsters, though effect size varied based on intervention length and delivery format

Report the Review

- Good research isn't just about doing the work; it's about clearly presenting it.
- Follow the PRISMA reporting guidelines, which help you include all critical elements of a systematic review.
- Structure your paper with:
  - Introduction – Define the research gap
  - Methods – Detail your search strategy, criteria, and tools
  - Results – Present study findings and PRISMA flow diagram
  - Discussion – Interpret results, limitations, and implications
  - Conclusion – Summarize takeaways
- Don't forget to include:
  - A search strategy appendix
  - A clear PRISMA diagram showing how many studies were included/excluded